

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe the working principle and basic components of hydraulic system. Also explain its different types in brief.
- Q.37 a) Describe the constructional details and working principles of Disc harrow.
b) What are different types of furrow openers?
- Q.38 What are different types of seed metering mechanisms in seed drill and planter?

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4th Semester/ Agri Subject : Farm Machinery & Implements-I

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The method of planting in which row to row and plant to plant distance is uniform
a) Dibbling b) Drilling
c) Broadcasting c) Check row planting
- Q.2 A method of correcting unwanted forces
a) Vibration b) Tension
c) Balancing d) Centrifugal Force
- Q.3 A machine used for sowing wheat without any extra tillage operation.
a) Seed drill b) Strip till drill
c) No till drill d) Zero till drill
- Q.4 It is a part of a machine which has been manufactured without the operation of assembly.
a) Element b) Link
c) Bearing d) Joint
- Q.5 The implement used for removing weeds
a) Hand hoe b) Plough
c) Ridger seeder d) Digger

- Q.6 An implement that cuts the soil to a shallow depth for smoothening and pulverizing the soil and control weeds.
 a) Plough b) Harrow
 c) Cultivator d) Ridger
- Q.7 An implement used for digging holes for tree plantation
 a) Seed drill b) Planter
 c) Post hole digger d) Digger
- Q.8 A plough used to cut and pulverize soil by impact forces by means of a number of rotary tines or knives.
 a) Rotary Plough b) Rotavator
 c) Subsoiler d) Ridger
- Q.9 A part of plough which slides against the furrow walls and give lateral stability in the plough
 a) Share b) Mouldboard
 c) Landside d) Frog
- Q.10 Tilt angle varies from
 a) 10 to 15° b) 15 to 25°
 c) 25 to 30° d) 30 to 35°

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 What is the use of three point linkage?
- Q.12 Name two types of bearings.
- Q.13 Give two differences between seed drill and planter.
- Q.14 Name any two types of tillage.
- Q.15 State the function of transplanter.
- Q.16 Give the function of paddy weeds.
- Q.17 State two limitations of mechanization.

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- Q.18 Name two types of joints used in agricultural machinery.
- Q.19 Name two types of plough accessories.
- Q.20 Define horizontal suction.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 How seeds drill is different from planter?
- Q.22 Give classification of farm implements according to hitching.
- Q.23 What are different methods of balancing farm machinery.
- Q.24 Describe different methods of sowing field crop.
- Q.25 Describe the working of trans planter.
- Q.26 Explain the routine maintenance of tillage machinery.
- Q.27 What are the advantages of using Rotavator over other implements?
- Q.28 What should be the conditions for maximum power transmission?
- Q.29 What is the scope of mechanization?
- Q.30 What are the different adjustments in disc plough?
- Q.31 State the harmful effects of vibrations in machines and their remedies.
- Q.32 Describe the working principle of zero tillage machinery.
- Q.33 What are different methods of power transmission in machines?
- Q.34 Compare disc plough and mouldboard plough.
- Q.35 Describe two types of couplings used in machinery.

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